

Zihao Liu

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EDUCATION

University of Edinburgh

Edinburgh, United Kingdom

Master's Degree in High Performance Computing

Sep. 2023 – Aug. 2024

- Member of TeamEPCC - Represented the university at the International Supercomputing Conference (ISC) Student Cluster Competition (SCC) in Germany (May 2024).
- **Grade:** Merit

University of Bristol

Bristol, United Kingdom

Bachelor's Degree in Electrical and Electronic Engineering

Sep. 2019 – Jun. 2023

- **Grade:** Second Class Honours (First Division).

WORK EXPERIENCES

Arm

Manchester, United Kingdom

RTL Design Engineer

Apr. 2026 – Present

- Working in Arm System Memory Management Unit (SMMU) team.
- Responsible for the design and implementation of the Bus Interface Unit in SMMU, functioning as a read/write buffer between external device and Translation Lookaside Buffer meanwhile handling the conversion between AMBA AXI and LTI transactions.
- Investigated and created detailed design report of reset domain crossing solutions for a new multi-reset pin feature.

Arm

Manchester, United Kingdom

Graduate Engineer (Rotation Program)

Sep. 2024 – Mar. 2026

RTL Design Rotation:

- Worked on the design of parity protected control/data register module, synthesized the design using cell libraries with different voltage thresholds and performed timing analysis to identify max frequency and timing bottlenecks.
- Developed a Python module usage analysis tool to scan the codebase, map instance parameter usage into Excel reports, and identify and eliminate redundant module parameters.
- Developed custom JasperGold linting rules using TCL to enforce coding standards and automatically flag non-compliant RTL patterns.

Formal Verification Rotation:

- Developed three formal verification testbenches from scratch for previously unverified SMMU blocks, discovering 14 critical functional bugs missed by dynamic verification.
- Accelerated designer debug time by independently root-causing failing formal properties and isolating the underlying RTL issue prior to hand-off.
- Maintained and stabilized legacy formal testbenches by pruning unreachable properties, resolving over-constraints, and developing abstraction models for complex control and data paths.
- Applied formal techniques including bounded coverage analysis, ProofCore/COI coverage, X-propagation checking, and proof-depth tuning.
- Integrated formal testbenches with dynamic regressions, debugging and resolving over-constrained properties to ensure formal models accurately reflect intended behavior.

Huawei Technologies Research & Development (UK) Ltd

Edinburgh, United Kingdom

Research Intern

Apr. 2024 – Aug. 2024

- Developed a Python-based Gem5 checkpointing tool to extract granular hardware simulation statistics (TLB/cache hits, latency, bandwidth) to drive CPU memory management research.
- Streamlined simulation workflows by generating checkpoints from simple Gem5 models that initializes client-server apps (Redis, MySQL), allowing researchers to quickly restore them on complex Gem5 models for deep, fine-grained analysis while running the apps.
- Automated and parallelized the restoration of multiple Gem5 simulations across diverse boot configurations.
- Built custom Linux root file systems, disk images, and kernels for QEMU and Gem5 environments.

Arm

Manchester, United Kingdom

Hardware Engineering Intern

Jul. 2023 – Mar. 2024

- Contributed to a Python-based register generation workflow that automatically generates the RTL and UVM

coverage files for register host blocks, which helps eliminating error-prone coding activities for register access RTL and verification. Designed four reusable SystemVerilog templates in the workflow to generate RTL and UVM code for 40+ register host blocks.

- Implemented a new feature where designers can easily specify register field guarding expressions in YAML format that can be integrated in the workflow for RTL and UVM generation.
- Identified and improved inefficiencies in the workflow, reducing workflow run time by at least 20%.
- Gained exposure to industry-standard best practices through daily technical design and verification meetings.

Huawei Technologies Research & Development (UK) Ltd

Cambridge, United Kingdom

Research Intern

Feb. 2023 – Jun. 2023

- Architected CI/CD pipelines from scratch using GitLab CI and Jenkins to automate regression testing.
- Improved computational resource efficiency by containerizing test environments with Docker and deploying them to Kubernetes clusters.
- Developed Jenkins pipelines for nightly and sanity regressions, utilizing plugins for automated cross-pipeline triggering and email alerts.
- Created a Python data visualization tool to parse LLVM test suite logs and generate graphical regression reports.

SKILLS & INTERESTS

Skills: English (Fluent), Chinese (Native), SystemVerilog, SVA, RTL Design, Formal Verification, QuestaSim, Cadence JasperGold, Python, Linux, Git, CI/CD

Interests: Badminton, Football, Volleyball, Hiking, Board Games, League of Legends